

Course Write-Up (for Courses other than External Degree Programmes)

Key items	PEIs' Inputs															
Course Title: 1. If it is 100% e-learning course, add "(e-learning)" at the end of the course title. 2. If conducted in Mandarin, add "(Mandarin)" at the end of the title. 3. Include prefix "(MC)" for stackable certificate courses.	Advanced Certificate in AI Security Analyst															
Course Developer	☒ Self-developed ☐ Externally-developed ☐ Jointly-developed <i>(Curriculum self-developed by Tertiary Infotech.)</i>															
Qualification to be Awarded upon Course Completion <i>Note: The name of qualification to be awarded should be the same as the course title.</i>	Advanced Certificate in AI Security Analyst															
Qualification to be Awarded by:	Tertiary Infotech Academy															
Brief Description of Course (including learning objectives for each module)	Brief description of course and learning outcomes: The Advanced Certificate in AI Security Analyst equips cybersecurity practitioners with the specialised skills to secure AI and machine learning (ML) systems, defend against AI-driven threats, leverage AI to enhance security operations, and govern AI responsibly in line with global regulatory frameworks. Upon completion, learners will be able to identify and analyse AI-specific threats, implement security controls across the AI lifecycle, automate and augment security operations using AI tooling, and apply AI governance, risk and compliance (GRC) practices. Each module is consists of 10 sessions (9 teaching + 1 assessment), conducted live online for 3 days a week, 7:00 PM – 10:00 PM (3 hours per session). : <table border="1"> <thead> <tr> <th>S/N</th><th>Module Title</th><th>Learning objective</th></tr> </thead> <tbody> <tr> <td>1</td><td>Module 1: Basic AI Concepts Related to Cybersecurity</td><td>This module develops foundational understanding of AI, machine learning, deep learning and NLP concepts as applied to cybersecurity, and the ability to recognise AI-driven threats such as adversarial attacks and malicious generative AI.</td></tr> <tr> <td>2</td><td>Module 2: Securing AI Systems</td><td>This module builds advanced skills to implement security controls that protect AI systems, data and models across on-premises, cloud and hybrid environments, and to mitigate adversarial risks targeting AI models and data pipelines.</td></tr> <tr> <td>3</td><td>Module 3: AI-Assisted Security</td><td>This module develops the ability to enhance threat detection and response using AI-driven tools, automate security workflows through event triage and alert correlation, and apply AI techniques to threat modelling and continuous monitoring.</td></tr> <tr> <td>4</td><td>Module 4: AI Governance, Risk, and Compliance</td><td>This module develops competency in applying AI governance frameworks and global regulatory requirements, integrating GRC practices throughout the AI lifecycle, and ensuring responsible AI use in alignment with GDPR and the NIST AI RMF.</td></tr> </tbody> </table>	S/N	Module Title	Learning objective	1	Module 1: Basic AI Concepts Related to Cybersecurity	This module develops foundational understanding of AI, machine learning, deep learning and NLP concepts as applied to cybersecurity, and the ability to recognise AI-driven threats such as adversarial attacks and malicious generative AI.	2	Module 2: Securing AI Systems	This module builds advanced skills to implement security controls that protect AI systems, data and models across on-premises, cloud and hybrid environments, and to mitigate adversarial risks targeting AI models and data pipelines.	3	Module 3: AI-Assisted Security	This module develops the ability to enhance threat detection and response using AI-driven tools, automate security workflows through event triage and alert correlation, and apply AI techniques to threat modelling and continuous monitoring.	4	Module 4: AI Governance, Risk, and Compliance	This module develops competency in applying AI governance frameworks and global regulatory requirements, integrating GRC practices throughout the AI lifecycle, and ensuring responsible AI use in alignment with GDPR and the NIST AI RMF.
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Target Audience	Primary Target Audience - IT and cybersecurity professionals who want to specialise in securing AI/ML systems (e.g., SOC analysts, security engineers, system/network administrators). - Cybersecurity practitioners seeking to use AI to enhance detection, response and security operations. - Fresh graduates from IT, computer science, data science or engineering															

	programmes who want career-ready AI security skills. • Mid-career professionals seeking a career switch into AI security or AI-focused cybersecurity roles. • GRC analysts, auditors and compliance officers who need to govern AI systems and meet AI regulatory requirements. • ML/data practitioners moving into security-focused roles who need to secure the models and pipelines they build. • Technical managers or team leads who need a strong AI security foundation to guide their teams or projects.																																
(For certificate and above qualifications only) – Potential Employment Opportunities upon Course Completion	Graduates of the Advanced Certificate in AI Security Analyst can pursue a range of roles in both the public and private sectors, including: • AI Security Analyst • AI/ML Security Engineer • Security Operations Center (SOC) Analyst (AI-augmented) • Cyber Threat Intelligence Analyst • AI Governance, Risk & Compliance (GRC) Analyst • AI Risk & Assurance Specialist • Machine Learning Security Specialist • Cybersecurity Analyst / Engineer • Incident Response Analyst • Cloud & AI Security Associate • Responsible AI / AI Compliance Officer These roles span industries such as finance, government, healthcare, technology and critical infrastructure, with opportunities for advancement into senior and managerial AI security positions.																																
Does the proposed course curriculum follow a foreign or international curriculum through full-time programme at the Primary, Secondary, and/or High School levels (grades 1-12), including self-developed curriculum?	☐ Yes ☒ No <i>If yes, please state the location of PEI's premise for the course to be conducted in: N/A</i>																																
Course Details	Mode of Delivery: ☐ Face-to-face ☐ Blended ☒ E-learning <i>(100% synchronous live virtual classes)</i> Duration: ~3.5 months (part-time) Class Frequency: 3 days per week x 3 hours per day (part-time), 7:00 PM – 10:00 PM Total Course Hours: 120 hours (part-time) <i>Computation for course hours:</i> <table><tr><th>S/N</th><th>Course component</th><th>Number of Hours</th><th>Remarks</th></tr><tr><td>1</td><td>Classroom Facilitation</td><td>N/A</td><td>Course is 100% e-learning (virtual)</td></tr><tr><td>2</td><td>Practicum</td><td>N/A</td><td></td></tr><tr><td>3</td><td>Practical</td><td>54</td><td>Hands-on labs in cloud sandbox / Jupyter / SIEM environments (delivered live online)</td></tr><tr><td>4</td><td>Synchronous e-learning</td><td>54</td><td>Live instructor-led virtual lectures & demonstrations</td></tr><tr><td>5</td><td>Asynchronous e-learning</td><td>N/A</td><td></td></tr><tr><td>6</td><td>Assessment</td><td>12</td><td>3-hour assessment after each of the 4 modules</td></tr><tr><td></td><td>Total Duration (in hours)</td><td>120</td><td>40 sessions x 3 hours (36 teaching + 4 assessment)</td></tr></table>	S/N	Course component	Number of Hours	Remarks	1	Classroom Facilitation	N/A	Course is 100% e-learning (virtual)	2	Practicum	N/A		3	Practical	54	Hands-on labs in cloud sandbox / Jupyter / SIEM environments (delivered live online)	4	Synchronous e-learning	54	Live instructor-led virtual lectures & demonstrations	5	Asynchronous e-learning	N/A		6	Assessment	12	3-hour assessment after each of the 4 modules		Total Duration (in hours)	120	40 sessions x 3 hours (36 teaching + 4 assessment)
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Minimum Entry Requirements	Age: 21 years old Language Proficiency: At least C6 for GCE “O” Level English Academic: At least C6 for GCE “O” Level in any 3 subjects <i>Recommended (not mandatory): basic IT/cybersecurity knowledge (e.g., CompTIA Security+ or equivalent) and basic Python/scripting familiarity.</i>																																
(For certificate qualifications only) – to be read in conjunction with SSG	Will the PEI issue modular certificate(s) that students can stack towards the award of the Certificate Qualification?																																

Advisory 3/2022	☒ YES ☐ NO <i>Note: Only "Terminal" Certificate Courses can be registered as Modular Courses that stack to a Certificate Qualification.</i> <table border="1" data-bbox="570 228 1507 453"> <thead> <tr> <th data-bbox="570 228 643 289">S/N</th><th data-bbox="643 228 1005 289">Title of certificate qualification</th><th data-bbox="1005 228 1507 289">Modules</th></tr> </thead> <tbody> <tr> <td data-bbox="570 289 643 453">1.</td><td data-bbox="643 289 1005 453">Advanced Certificate in AI Security Analyst</td><td data-bbox="1005 289 1507 453"> 1. Basic AI Concepts Related to Cybersecurity 2. Securing AI Systems 3. AI-Assisted Security 4. AI Governance, Risk, and Compliance </td></tr> </tbody> </table> Maximum time allowed to complete the modular courses for award of the certificate qualification: ~3.5 months (part-time). Terms and conditions to be met to be awarded the Certificate: To be awarded the certificate, participants must: • Achieve a pass in all required assessments (one per module), and • Maintain a minimum attendance of 75% throughout the course.	S/N	Title of certificate qualification	Modules	1.	Advanced Certificate in AI Security Analyst	1. Basic AI Concepts Related to Cybersecurity 2. Securing AI Systems 3. AI-Assisted Security 4. AI Governance, Risk, and Compliance
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Articulation Pathway	Is the proposed course a pathway programme? ☐ YES ☒ NO						
Information on External Course Developer <i>(for externally developed courses only)</i>	Not applicable — the course is self-developed by Tertiary Infotech Academy.						
Course Accreditation	Is the proposed course accredited by any external organisation? ☐ YES ☒ NO						
Association, Collaboration or Affiliation	Is there any other form of association, collaboration or affiliation with any other organisation or persons, either local or foreign, in respect of this course? ☐ YES ☒ NO						
Courses with Industrial Attachment	Does the proposed course have industrial attachment? ☐ YES ☒ NO						
Other Information, if applicable	N.A.						

I, **Dr. ANG CHEW HOE**, (NRIC/Passport no. S1808997A) manager of **TERTIARY INFOTECH ACADEMY PTE. LTD.** declare that:

- (a) the course has been approved by the Academic Board;
- (b) the Academic Board has approved the deployment of teachers for the course in accordance with Regulation 26 of Private Education Regulations;
- (c) the assessment methods and procedures for the course have been approved by the Examination Board; and
- (d) all information provided in this application is true and accurate to my knowledge.

Alfred Ang

Date: 12 Jun 2026

(Signature of manager & date)